3

Types Of Clouds And Cloud Providers

3.1 Introduction

The previous chapter explains the concept of elastic computing. The chapter describes how virtualization provides the basic technology that enables a provider to increase or decrease the servers leased by a customer quickly. The chapter also describes the business models used in the cloud industry.

This chapter continues the discussion of the business models. The chapter explains how a large company can use cloud technology internally and how public cloud providers arose to sell cloud services.

3.2 Private And Public Clouds

Cloud computing technologies have been used in two fundamental ways. To describe the two, industry uses the terms:

- Private Cloud – an internal cloud used only by one organization
- Public Cloud – a commercial service used by multiple customers

The next sections examine each of the types along with underlying motivations. Later sections explain how organizations use combinations and variations.

3.3 Private Cloud

We use the term *private cloud* to describe a cloud data center that is owned and operated by an organization and restricted to the organization's computing. Chapter 1 points out that an organization can reduce costs by consolidating all of its servers by placing them in racks in a data center. The organization can further reduce costs by employing cloud technologies. To understand why, suppose three servers have been collected from department A and five from department B. If the servers remain dedicated to departments, a given server may be underutilized or overutilized. More important, if the computing demands of departments vary, one department's servers may be overloaded, while another department's servers are underutilized. When a server becomes oversubscribed, critical business functions can suffer. From an economic point of view, underutilized servers represent unwarranted cost, and the inability to expand facilities quickly represents a risk to business. Of course, an organization must have sufficient resources to handle the peak load, but at other times, the organization can use the elastic approach described in Chapter 2. Instead of physical servers, the approach allows departments to create and assign virtualized servers to balance the load across all physical servers. The point is:

> *When an organization uses cloud technology on an internal data center, we say that the organization owns and operates a* private cloud*. A private cloud can avoid underutilized as well as oversubscribed resources by spreading computing across all physical servers.*

3.4 Public Cloud

A company that offers cloud computing to its customers is said to operate a *public cloud*, and the company is known as a *public cloud provider*. For an organization, subscribing to a public cloud constitutes the chief alternative to operating a private cloud internally. When using a public cloud, an organization leases services, including virtualized servers, and then uses the leased facilities to perform computation. We can summarize:

> *A company that offers cloud services to customers is known as a* public cloud provider*, and the cloud facility that such a company operates is known as a* public cloud facility.

When using a public cloud, an organization must choose a type of service, as Chapter 2 describes. A medium or large size enterprise customer is likely to choose an IaaS service that offers a set of virtualized servers, an amount of block storage, and Internet connectivity. A large customer may also choose to pay for additional services, such as enhanced data backup, a specialized security service, and database administra-

tion. Customers who intend to create and deploy their own apps, may subscribe to a PaaS service that makes it easy to create software and run apps in the provider's cloud. A smaller public cloud customer may choose to subscribe to specific SaaS services, such as a service that runs a web site for the customer or software used to prepare documents.

3.5 The Advantages Of Public Cloud

Why would a corporation choose to use a public cloud provider? Cloud providers advertise three main advantages:

- Economic – much lower cost than a private cloud
- Expertise – access to a staff with expertise on many topics
- Advanced services – offerings not available elsewhere

The economic advantage. Because they serve multiple tenants, cloud providers operate data centers that are much larger than private cloud data centers owned and operated by individual organizations. Therefore, public cloud providers benefit from economy of scale: they can spread costs among more customers, and can negotiate larger discounts on commodity servers and network equipment. For example, instead of conventional (i.e., expensive) network switches, providers use what the industry calls *whitebox switches*. The disadvantage of a whitebox switch is that it only includes basic packet forwarding hardware, and does not provide software that an IT manager can use to configure and control the switch. To operate their whitebox switches, cloud providers employ a technology known as *Software Defined Networking* (*SDN*). A manager connects a computer to each of the switches, and runs SDN control software on the computer. The manager can then interact with the control software to specify the desired configuration, and the control software installs forwarding tables in the whitebox switches that fulfill the manager's specifications.

Discounts on equipment and the use of whitebox network switches means that a public cloud provider spends less on each piece of equipment than a company does when they operate a private data center. In addition, a public cloud provider shares IT staff expertise across all tenants, the cost of staff for a given tenant is less than the cost of staff for a private data center. The savings can be substantial — an employee at Azure claims they can deploy a server for 25% of the cost of a private cloud deployment. The bottom line is:

> *By sharing IT expertise across multiple tenants, negotiating large discounts on commodity servers, and using SDN to control whitebox network switches, a public cloud provider can operate a cloud data center for substantially less than it costs to operate a private cloud data center.*

The expertise advantage. Because it is much larger than the IT department in a single company, a public cloud provider can afford to maintain a staff with specialized expertise. Thus, customers have access to expertise on a much wider range of topics than a local IT staff can provide.

In addition to traditional data center software, such as operating systems, relational database services, and technologies for virtualized servers, cloud providers now include expertise on the broad topic of *Artificial Intelligence* (*AI*) and a technology that AI uses, *machine learning* (*ML*) software.

The advanced services advantage. Initially, providers investigated AI/ML as a way to automate the management of data center hardware and software. For example, by using ML to monitor network traffic, a provider can detect anomalies, such as misrouted traffic or unexplained spikes in load. Industry uses the term *AIops* to refer to the use of AI/ML to automate data center operations.

More recently, cloud providers have begun offering AIops as a service to customers. In addition to monitoring and automating the deployment and operation of customer apps, providers offer advanced services that use AI/ML software to analyze business data. For example, some AI/ML offerings help companies understand sales trends and can help spot repeat customers.

In addition to AI-based services, providers offer a variety of other advanced services. For example, providers offer services that update a customer's operating systems and other standard software whenever the vendor releases an update. Major cloud providers also offer platforms that allow a customer to build and deploy new apps without understanding the details of the underlying system. In particular, a software engineer can focus on building an app without learning how to create and replicate virtualized servers, establish network connectivity, or handle remote storage. One of the advantages of using such a platform lies in its safety — because the service details deployment details, a customer is less likely to make a mistake that either causes the app to misbehave or incurs excessive cost by consuming data center resources unnecessarily. The point is:

> *Advanced services offered by providers give customers access to AI/ML software and allow customers to create and deploy applications quickly, easily, and safely.*

3.6 Provider Lock-In

Public cloud providers compete to attract new customers and entice existing customers to increase their monthly subscription. One approach focuses on offering specialized services that are not available elsewhere. For example, a provider may offer customers a network management service that allows each customer to monitor and control the network used internally (i.e., between the customer's virtualized servers) and externally (i.e., between a customer's virtualized servers and the rest of the Internet). Because it interacts with the provider's network control system, such a service is inherently linked to the provider.

In addition to advanced services that link to the provider's infrastructure, cloud providers usually offer a *migration service* that makes it easy for a corporate customer to move their computing into the provider's public cloud. To entice customers to switch from another provider, a provider may also offer *cloud-to-cloud migration services*.

To retain customers, providers try to make it difficult for a customer to move to another provider. For example, a provider may offer free or nearly free services that makes it especially easy for a customer to perform routine tasks, such as generating monthly activity summaries, analyzing which of a customer's web pages attract the most attention, or testing new apps in a safe, isolated environment. The catch arises because the service only runs on the provider's infrastructure — if a customer transfers to another provider, the customer will no longer be able to use the service. Even if another provider offers a similar service, the details will change, and a customer must retrain staff for the new system. Industry uses the term *lock-in* to refer to incentives and obstacles that a provider uses to discourage customers from transferring to another provider.

We can summarize:

> *Industry uses the term* lock-in *to refer to the practice of using enticements and obstacles that make it inconvenient or expensive for customers to move to another cloud provider.*

3.7 The Advantages Of Private Cloud

It may seem that the advantages of public cloud — lower cost, access to specialized expertise, and advanced services — will drive all companies to use public cloud services. However, private cloud offers advantages:

- Retention of control and visibility
- Reduced latency with on-premises facilities
- Insurance against future rate hikes

Retention of control and visibility. A public cloud customer does not have access to the underlying infrastructure. If a problem arises, a customer can only report the visible symptoms, but cannot perform root cause analysis because only the cloud provider can examine the network switches and physical servers.

For organizations in a regulated industry, regulations may require the organization to control the placement and transmission of data as well as the hardware and software used. A private cloud makes it possible for such an organization to comply with the rules.

Reduced latency with on-premises facilities. Because they are located within a company, private cloud facilities are said to be *on-premises* (*on-prem*). The delay between employees and a private cloud data center can be significantly lower than the delay to a public cloud, especially if an organization is geographically distant from a public cloud site. An organization with multiple sites may be able to further reduce la-

tency by placing a private cloud data center at each site. If computation and communication remains primarily within a site, the improvement can be noticeable.

Insurance against future rate hikes. Provider lock-in presents a long-term liability for a company. As time goes by, a customer adopts more of the provider's services, and the cost of changing providers increases. A provider can raise rates without fear of losing customers. The point is:

> *Although public cloud seems to offer several compelling advantages, private cloud gives a company more control, can reduce latency, and guards against future rate hikes by a provider.*

3.8 Hybrid Cloud

Instead of choosing between public and private clouds, some organizations opt for a compromise that offers some of the advantages of each. Known as a *hybrid cloud*, the compromise means an organization uses a public cloud provider for some computing, and runs a private cloud for the rest. The balance between the two depends on the organization's needs as well as cost. Examples of why an organization might adopt a hybrid cloud approach include:

- Control when needed
- Computation overflow

Control when needed. We said that some organizations must comply with regulations. For example, a government contractor may need to store classified data on servers with controlled access. Using a hybrid approach allows such a company to enforce restricted access for its private cloud, and reduce overall costs by pushing other data and computations to the public cloud.

Computational overflow. Imagine an organization that uses a private cloud for most of their computing needs. During the organization's peak business season, the private cloud may have insufficient resources to handle the load. Rather than add additional servers, which will remain underutilized most of the time, the organization can send some of the computation to a public cloud during the peak period. To make overflow processing convenient, public cloud providers offer software that helps automate the process. In essence, the software makes the customer's private cloud compatible with the provider's public cloud, meaning that a customer can move overflow from their private cloud to the public cloud with little or no effort. We can summarize:

> *Using a hybrid cloud allows a company to enforce restrictions on data and computations in their private cloud and reduce costs by pushing the rest to a public cloud; the hybrid approach can also be used to handle overflow.*

Of course, the hybrid approach does have some disadvantages. Consider, for example, security management. An organization has complete control over their private cloud and the security systems being used, but will depend on a provider's security systems when using a public cloud. Even if the two systems offer the same functionality, the IT staff must learn to configure and operate two systems.

3.9 Multi-Cloud

Because provider lock-in represents a potential long-term liability, large organizations try to avoid dependence on a single public cloud provider. To take advantage of public cloud services while avoiding lock-in, large organizations adopt an approach that has become known as *multi-cloud*, which means an organization becomes a customer of more than one public cloud provider. The division of computation among providers depends on the structure of the organization and its IT needs. As an example, consider a large corporation with business units. It may be possible to assign some business units to one cloud provider and other business units to another.

Although it avoids lock-in, using multiple cloud providers introduces challenges. It may not be easy, for example, to switch computation or data from one cloud provider to another. Services available from one provider may not match the services available from another. Even if a service is available from multiple providers, the organization may need to write specialized software either to translate from one system to another, or to combine the output from the two providers to generate data for the entire organization.

3.10 Hyperscalers

Industry uses the term *hyperscale* to refer to the largest cloud computing facilities, and the term *hyperscalers* to refer to the companies that own and operate such facilities. Social media and search companies, such as Facebook, Twitter, and Google were among the earliest to build giant data centers to support their respective businesses. Because such companies focus on consumers, early definitions of hyperscale measured the number of simultaneous users data centers could handle. The shift to more general-purpose cloud computing has made the measure of users irrelevant, and the definition has shifted to a measurement of how much a company spends on data center infrastructure. The numbers are huge. By 2018, Google, Amazon, Microsoft, Facebook, and Alibaba were spending about $11B per year just on new servers.

In terms of the public cloud services used by large enterprise organizations, three hyperscalers stand out:

- Amazon's AWS (Amazon Web Services)
- Microsoft's Azure Cloud
- Google' GCP (Google Cloud Platform)

Growth of the cloud industry has been rapid, especially for the top two providers. Initially, Amazon built data centers to support its online retail business. In 2006, before most enterprises were even aware of general cloud computing, Amazon launched AWS. In early 2010, Microsoft released *Windows Azure*, and four years later renamed it *Azure Cloud*. In 2012, cloud services formed an insignificant percentage of revenue for both Amazon and Microsoft. By 2018, just six years later, Microsoft reported that cloud services generated $32.2B or 29% of total revenue. By 2018, AWS generated $25.7B for Amazon or 11% of total revenue.

Hyperscalers are shifting spending to match a shift in services. In 2018, for example, Microsoft revealed it had invested $20B to create its Azure Cloud platform. Now that the data centers have been built, hyperscalers are shifting investment toward AI/ML technologies that will allow them to automate management of their cloud infrastructure and offer advanced services to customers.

3.11 Summary

Cloud computing can be divided into two basic types: public and private. An organization that owns and operates a data center for internal use is said to operate a private cloud, and an organization that owns and operates a data center that external customers use for their computing is said to operate a public cloud.

Because both private and public clouds have advantages, some organizations choose a hybrid approach in which some computing is performed in the organization's private cloud, and other computation is performed in a public cloud. An organization can also adopt a multi-cloud approach in which the organization uses multiple public cloud providers.

Companies that own and operate the largest data centers are known as hyperscalers. Among public hyperscale cloud providers, Amazon's AWS, Microsoft's Azure Cloud, and Google's GCP have attracted enterprise customers.